

Student ID:

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Quiz no. 4

Winter Semester 2024-2025

DURATION: 20 MIN

FULL MARKS: 20

CSE 4107: Structured Programming I

Figures in the right margin indicate marks.

1. Find the output of the following program:

```
#include <stdio.h>

int main() {
    char s1[20] = "network";
    char s2[20] = "security";
    int i, j = 0;

    for (i = 0; s1[i] != '\0'; i++) {
        if (i % 2 == 0) {
            s1[i] = s1[i] - 32;
            s2[j] = s1[i];
            j++;
        } else {
            s2[j] = s2[i];
            j++;
        }
    }
    s2[j] = '\0';

    printf("%s\n", s1);
    printf("%s\n", s2);

    return 0;
}
```

8
(CO1)
(PO1)

2. A telecom company records customer call data. Each record contains:

- User ID (integer)
- Call Type (character; 'L' for Local, 'I' for International)
- Duration of call (double; in minutes)
- Cost per minute (float)

12
(CO3)
(PO1)

Write a function which takes two parameters: an array of call records, and the number of records. The function should return the total cost of all International calls. The function is declared as follows:

```
double International_cost(struct Record records[], int
                           num_records)
```